Michael Tin

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Education

University of California, Riverside (UCR) — Apr 2026 – Present
Master of Science in Computer Engineering — *Riverside, CA*

University of California, Riverside (UCR) — Sep 2021 – Jun 2025
Bachelor of Science in Computer Engineering — *Riverside, CA*

Skills

Languages: Python, Java, C, C++, C#, Rust, JavaScript, TypeScript, SQL, MATLAB, Verilog, VHDL, Assembly
Frameworks: React, React Native, Next.js, Express.js, Angular, Flutter, Electron, Tauri, PyTorch, TensorFlow
Developer Tools: Git, GitHub Actions, Docker, Kubernetes, PostgreSQL, MongoDB, WSL, CMake, GDB, QEMU
EDA and Simulation Tools: PSpice, Capture CIS, Vivado, ModelSim, Synopsys Custom Compiler, GTKWave

Projects

Tally | *Next.js, TypeScript, React, Tailwind CSS* — Apr 2026 – Jun 2026
- Developed a real-time tournament scheduling dashboard for FIRST LEGO League (FLL) events, integrating with Google Sheets to fetch live schedule updates, team base locations, and upcoming matches.
- Engineered a Tournament Schedule Generator to expedite event administration, supporting customizable parameters, dynamic grid/list viewing modes, and export capabilities to XLSX, PDF, and PNG formats.

Scansheet | *Rust, Tauri, JavaScript, React* — Sep 2025 – Jun 2026
- Developed a full-stack attendance management system that eliminates manual entry errors through an automated QR code scanning infrastructure, reducing administrative overhead in educational settings.
- Engineered comprehensive data portability tools to streamline data management, including bulk CSV imports, date/category filtering, customizable PDF reports, and custom badge generation for users.

XV6 Kernel Optimization | *C, Assembly, RISC-V* — Jan 2025 – Mar 2025
- Engineered advanced system calls and symbolic links for the XV6 kernel, expanding core functionality with file system traversal, inter-process communication, and optimized I/O operations.
- Implemented dynamic page fault management techniques, including lazy page allocation, Copy-on-Write (COW) fork, and memory mapping, to streamline program execution while minimizing system resource consumption.

Experience

ELOP Tutor — Aug 2025 – Present
Riverside STEM Academy — *Riverside, CA*
- Direct daily technical operations for Science Olympiad, Math Olympiad, and FIRST Lego League Challenge, managing scheduling, resource allocation, and student engagement for over 100 participants.
- Enforce strict laboratory safety standards and equipment handling protocols to guarantee a compliant, secure learning environment during hands-on fabrication and electronics testing initiatives.

Director — Oct 2022 – Jun 2025
Change in Scientific Importance for Youth (Delta SIFY) — *Riverside, CA*
- Founded and led the Highlander Invitational, UCR's annual Science Olympiad competition, growing it into the Inland Empire's largest STEM tournament and reaching over 2,000 students across three years.
- Directed the Science Olympiad Coaching Program at Riverside STEM Academy, recruiting, onboarding, and managing the weekly schedules of 25 volunteers to mentor a group of 50 middle school students.
- Implemented robust training and quality assurance protocols for a team of 300 volunteers, overseeing the efficient execution of logistics and preparation across all program initiatives.

Technology Director — Sep 2022 – Jun 2025
The Highlander Newspaper — *Riverside, CA*
- Engineered hardware and network infrastructure for UCR's student-run newspaper, providing on-demand technical support and workflow training to ensure minimal downtime during weekly production.
- Devised a website optimization strategy centered on content delivery optimization, reducing website latency by 82.8% and growing readership by 73.3% to 40,000 monthly average users (MAUs).